

Activity 5: Enterprise SDLC Workflow Simulation with AI

Purpose

Learn how AI can support developers throughout the Software Development Life Cycle (SDLC), from understanding business requirements to preparing an application for release.

Pre-requisites

- Completion of Activities 1–4
 - Working SmartBank EMI Calculator application
 - VS Code with Cline configured
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Estimated Duration

45 Minutes

Scenario

The SmartBank EMI Calculator project has successfully completed development. The Project Manager has requested a retrospective to understand how AI contributed throughout the project lifecycle and where it added the most value.

As a developer, your responsibility is to use AI to review each phase of the SDLC, summarize key activities, identify benefits, and recommend best practices for future projects.

Note: The objective of this activity is to understand AI's role across the complete SDLC. This is a simulation based on the work completed during the previous activities.

Tasks

Task 1: Map AI to the SDLC

Identify how AI supported each stage of the project lifecycle.

Suggested Prompt

Act as a Technical Lead.

Review the SmartBank EMI Calculator project completed during this training.

Map the use of AI across the Software Development Life Cycle.

For each SDLC phase explain:

- Objective
- How AI assisted
- Benefits achieved
- Developer responsibilities

Cover the following phases:

- Requirements Analysis
- Development
- Refactoring
- Testing
- API Integration
- Code Review
- Release Preparation
- Maintenance

Present the information in a structured table.

Task 2: Evaluate AI Adoption

Review the strengths and limitations of using AI during development.

Suggested Prompt

Based on the SmartBank EMI Calculator project, evaluate the use of AI during software development.

Identify:

- Tasks where AI added the most value
- Tasks requiring developer expertise
- Situations where AI responses should be verified
- Limitations of AI-generated solutions

Summarize your findings with practical recommendations for development teams.

Task 3: Recommend AI Best Practices

Generate guidelines for developers who use AI in enterprise projects.

Suggested Prompt

Prepare a set of best practices for developers using AI coding assistants in enterprise software projects.

Include recommendations for:

- Writing effective prompts
- Reviewing AI-generated code
- Validating AI responses
- Protecting sensitive information
- Maintaining code quality
- Collaborating with team members

Present the guidelines as a checklist.

Task 4: Prepare a Project Retrospective

Summarize the complete project journey.

Suggested Prompt

Prepare a project retrospective for the SmartBank EMI Calculator.

Include:

- Project objective
- Key activities completed
- AI-assisted tasks
- Challenges encountered
- Lessons learned
- Recommendations for future projects

Write the retrospective in a professional format suitable for sharing with the project team.

Expected Deliverables

By the end of this activity, you should have:

- AI and SDLC mapping
- AI adoption assessment
- Enterprise AI best practices checklist
- Project retrospective report